



Batterjee Medical College
For Science and Technology

Parathyroid Disorders

Internal Medicine Department

Knowledge

- Relate pathophysiological basics and etiology of parathyroid disorders
- Describe epidemiology, manifestations, complications and management of hyperparathyroidism
- Discuss calcium homeostasis
- Describe causes, manifestations of hypercalcemia

Skills

- Interpret clinical manifestations and investigations of parathyroid disorders to formulate reasonable diagnosis.
- Apply evidence-based management plan for hyperparathyroidism
- Distinguish patients in emergency situations to formulate a comprehensive management plan; ensuring keeping patients in normal homeostasis

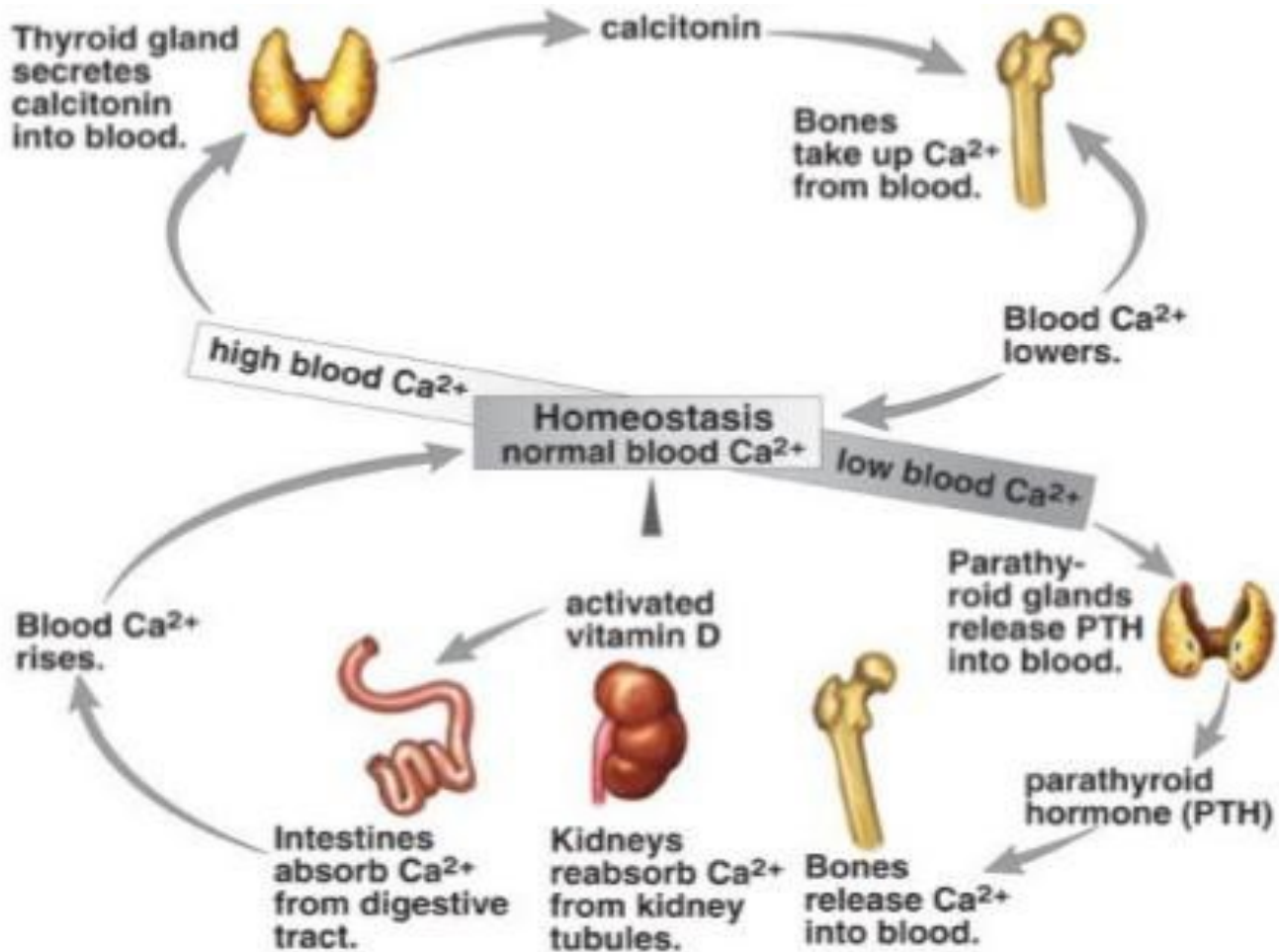
Main Topics

- Calcium homeostasis (parathyroid hormone and calcitonin regulation)
- Hyperparathyroidism (primary, secondary, tertiary, and ectopic causes; clinical manifestations; osteitis fibrosa cystica)
- Investigations and radiological findings in hyperparathyroidism
- Treatment of hyperparathyroidism (surgical management, hungry bone syndrome, treatment of severe hypercalcaemia)
- Causes of hypercalcaemia (endocrinal, malignant, iatrogenic, miscellaneous)
- Causes of generalized bone disease (renal osteodystrophy, osteomalacia, osteoporosis, Paget's disease, McCune-Albright syndrome, osteoblastoma)



Calcium Hemostasis

Physiological background on calcium regulation



- Serum Ca & P levels are controlled by certain hormones :
 - a. Parathormone.
 - b. Calcitonin.
 - c. Active vitamin D.
- Ca regulation involves 3 sites : bone, intestine & kidney.

	<i>Parathormone*</i>		<i>vitamin D</i>		<i>Calcitonin</i>	
Absorption (intestine)	↑ Ca	↓ P	↑ Ca	↑ P	↓ Ca	
Reabsorption (kidney)	↑ Ca	↓ P	↑ Ca	↑ P	↓ Ca	↓ P
Resorption (bone)	↑ Ca	↑ P	↑ Ca	↑ P	↓ Ca	↓ P
Blood (net result)	↑ Ca	↓ P	↑ Ca	↑ P	↓ Ca	↓ P

*PTH acts directly on bone & kidney & indirectly on intestine (through activation of vit.D).

Assessment

- Normal Ca level : 8.4 – 10.2 mg%.

A. Ionized (active) : 50%

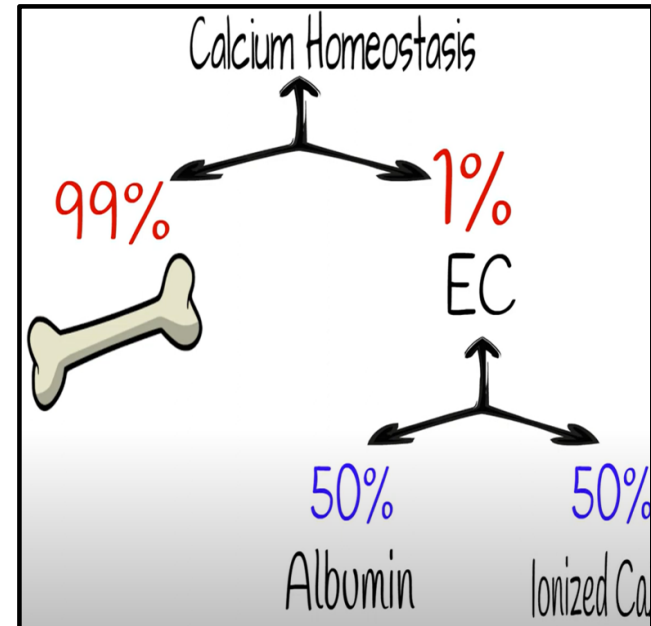
B. Non-ionized (reserve) : 50%

- 40% bound to albumin. - 10% any other bond (Ca carbonate, phosphate)

Liver cirrhosis & nephrotic syndrome → ↓ albumin → hypocalcaemia ,
but no tetany because ionized Ca is still normal.

- Regulation of Ca :

- Ca level is closely affected & related to P [n. 3 - 4.5 mg%]
- Notice that $Ca \times P = \text{constant}$ [40].

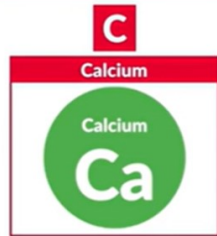


• **Total serum Calcium level** (8.5-10.5 mg/dL) after correction = (Correction of serum Ca for hypoalbuminemia **(Corrected by the level of albumin)**).

1- By adding 0.8 mg/dl of Ca for each 1 mg/dl of albumin reduction below 4 mg/dl.

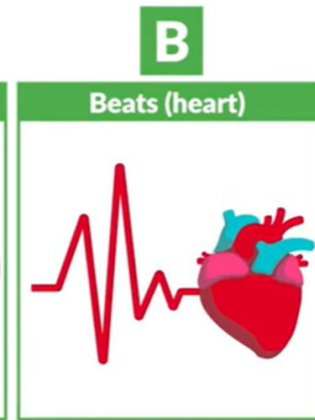
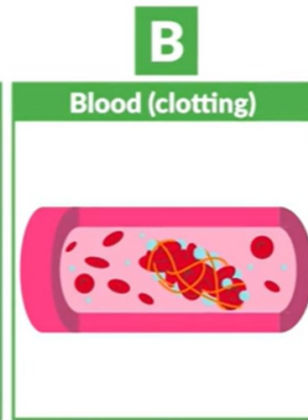
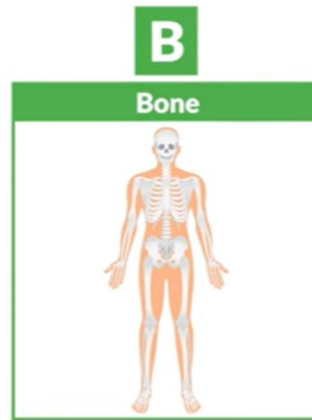
Corrected Ca = serum Ca + 0.8 x (Normal albumin - Pts' albumin)

Functions of Calcium



Keeps the
3 B's Strong

- B** - Bone
- B** - Blood (clotting)
- B** - Beats (heart)



Total serum Calcium level (8.5-10.5 mg/dL)

Serum ionized Ca < 4.5 mg/dL (Normal: 4.65-5.25 mg/dL).

Hyperparathyroidism (HPT)

‘Excess PTH Secretion’

1.Primary

- Adenoma (90 %).
- Hyperplasia.
- Carcinoma.
- Part of MEN I or II.

3.Tertiary hyperparathyroidism:

Autonomous PTH secretion superimposed on previous 2ry hyperparathyroidism.

2.Secondary:

In conditions with tendency to low serum Ca:

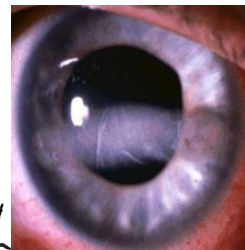
- Chronic renal failure " renal osteodystrophy “.
- Malabsorption syndrome and after gastrectomy.

4.Ectopic PTH secretion:

- Very rare & occurs in :
 - o Bronchogenic carcinoma.
 - o Hypernephroma.

Clinical Picture

Hypercalcaemia Causes metastatic calcification



Neuro-muscular & articular

- Fatigue, apathy, psychosis
- Pseudogout [Ca deposition in tendons & joint capsules]
- Muscular hypotonia [myopathy] → Waddling gait

Eye manifestations

- Ca may be deposited in 1 Conjunctiva & lateral edges of 1 Cornea [band Keratopathy]

Cardiovascular manif.

- Hypertension [due to ↑ vascular reactivity & ↑ renin]
- Hypotension may occur in severe cases due to dehydration.
- Arrhythmia [↓ cardiac arrest in systole]

Nodule in 1 neck

Skin manifest.

- Dry & may be Pruritis

Skeletal manif.

- Osteomalacia, osteoporosis, bone cysts

"Osteitis fibrosa cystica" [Von Recklinghausen's disease]

↓
pathological fracture

↓
[It heals rapidly due to hypercalcaemia]

Gastro-Intestinal manifestations

- Anorexia, nausea, vomiting & intestinal atony → Constipation.
- Peptic ulcer [in 15% of cases] → suggest Zollinger Ellison Syndrome
- Cholelithiasis, pancreatitis ← Ca ++ trypsinogen
Ca ppt in alkaline medium & panc. duct obst.

Renal Manifestations

- Excessive Ca diuresis → Polyuria & polydipsia [Ca diabetes]
- Recurrent bilat. Kid-stones & metastatic calcification [nephrocalcinosis]
- renal colic
- haematuria
- hydronephrosis
- Infection
- renal failure [if untreated]

Disease of bone, renal stone, abdominal groans & psychic moans

i. **Bone :** (Ostitis fibrosa cystica)

1. pain.
2. pathological fracture.

ii. **Renal :**

1. Repeated **stones**.
2. Polyuria (Ca diabetes) & polydipsia.
3. Nephrocalcinosis & renal failure may occur.

1. **CNS :**

- a. Drowsiness.
- b. Depression.

2. **GIT :**

- a. Peptic ulcer & Pancreatitis (**abdominal pain**).
- b. Constipation.

3. **CVS :**

- a. ECG : short Q-T interval & arrhythmia.
- b. Hypertension.

4. **Skin :**

- a. Dry.
- b. Itching.

Parathormone also tends to lower the blood phosphorus level.

Excess parathormone can result in markedly elevated levels of serum calcium, a potentially life-threatening situation.

When the product of serum calcium and serum phosphorus (calcium $\times$ phosphorus) rises, calcium phosphate may precipitate in various organs of the body and cause tissue calcification.

Clinical pearl

The majority of patients are asymptomatic.

Student pearl

"Stones, bones, abdominal groans, thrones, and psychiatric overtones!"

USMLE-style Question

Key learnings: signs of hypercalcemia

- Painful bones
- Renal stones
- Abdominal groans
- Sitting on the throne
(polyuria, constipation)
- & Psychiatric overtones.



Osteitis Fibrosa Cystica

- Classic localised bone lesion of hyperparathyroidism. Bone is lysed by osteoclasts driven by elevated PTH. Marrow replaced by highly vascularised fibrous tissue. Stress on weakened bone causes haemorrhage and cyst formation.
- Old term for this lesion was “brown tumour”. Colour due to massive haemosiderin deposition
- Typically found in jaw and long bones and may cause pathological fractures
- Can be distinguished from other giant cell tumours of bone by estimation of serum Ca.



1. Serum Ca: $\uparrow$ (N=9- 10.5 mg %).

2. Serum phosphate: $\downarrow$ (except in the presence of renal failure)
(N= 2.5- 4 mg %).

3. Serum alkaline phosphatase: $\uparrow$ in patients with bone disease.

4. Plasma PTH: $\uparrow$ (urinary C-AMP $\uparrow$ in hyperparathyroidism but not in other causes of hypercalcaemia).

5. Hypercalciuria: $\rightarrow$ 300 mg / 24 hr. on a low Ca diet is consistent with hyperparathyroidism (but not diagnostic).

6. Radiological finding :

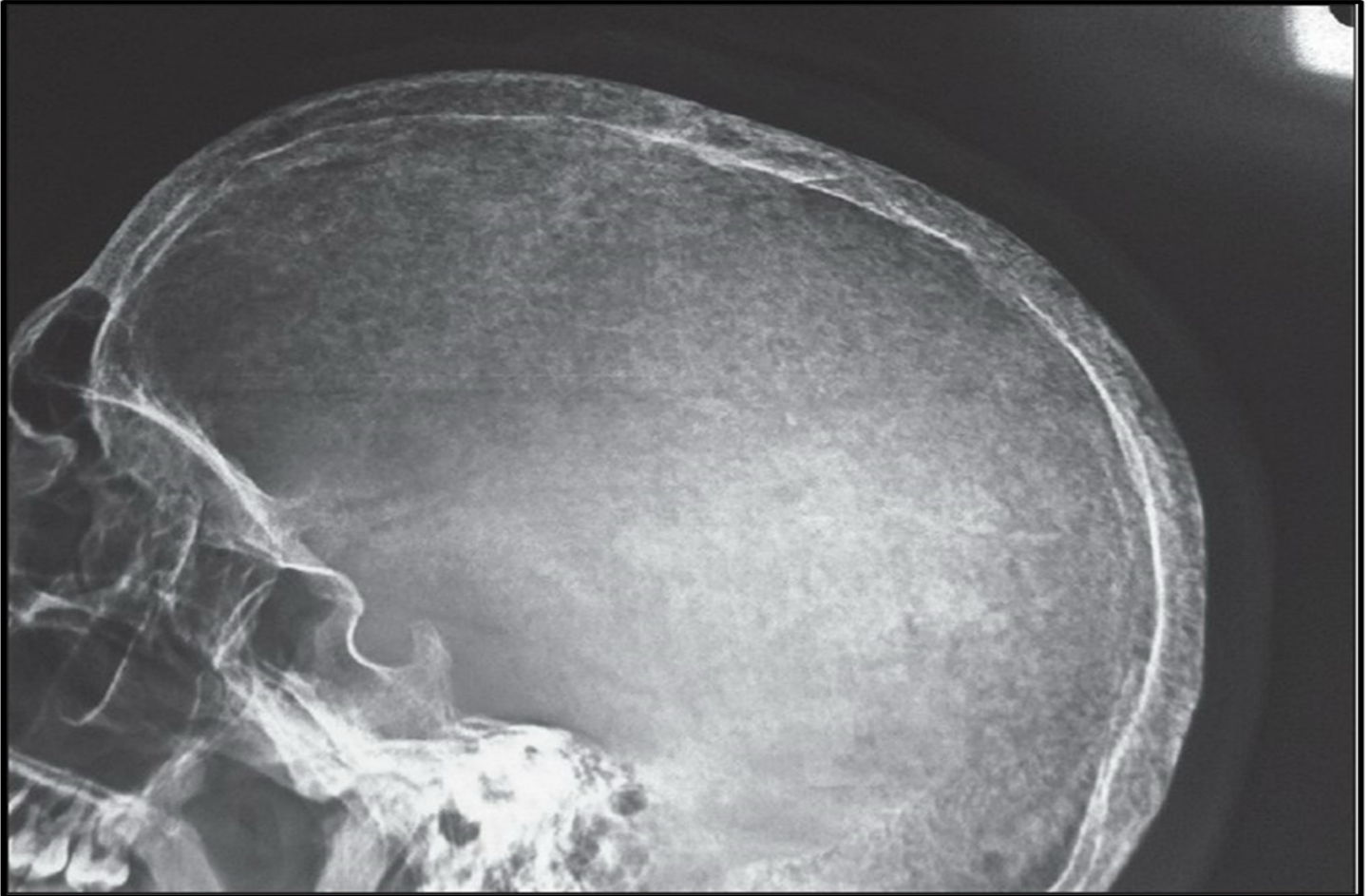
A. Kidneys may show: stones & nephrocalcinosis.

B. Skeletal changes:

- Osteitis fibrosa cystica (generalisata)
- Tiny, punched out lesions in the skull → salt & pepper appearance.
- Diffuse softening of the spine & hence indentation of the bodies of the vertebra by the intervertebral disc → cod fish spine
- Milk man fracture or looser zone :(Pseudofracture) A tongue of radio-translucency extending into the bone from the surface, mostly seen in the upper end of femur & humerus (due to osteomalacia).

C. Soft tissue calcification

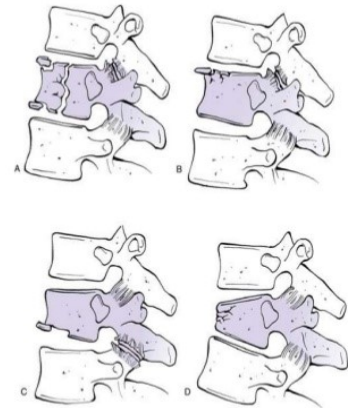






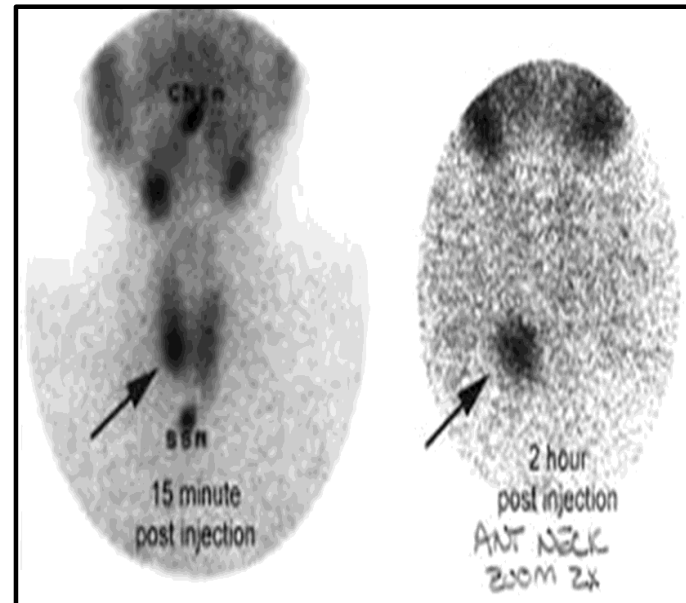
4 subtypes on basis of
end plate involvement

- # of both end plates
- # of superior end plate
- # of inferior end plate
- both end plates intact



7. For localization of the site of adenoma :

- Arteriography.
- Ultrasonography & CT scan of the neck.
- Technitium thallium scanning:
Technetium visualizes only the thyroid while thallium visualizes both thyroid & parathyroid.



Clinical pearl

Parathyroid imaging is used for surgical planning but is not indicated for initial diagnosis.

8. Steroid suppression test (Dent test): hypercalcaemia due to hyperparathyroidism doesn't suppress with hydrocortisone unlike other hypercalcaemic states.

1.Surgical

- **Indications:** symptomatic patients, and asymptomatic patients who fulfill any of the following criteria
 - Age < 50 years
 - Hypercalcemia: serum calcium level > 1 mg/dL above the ULN
 - Renal involvement
 - Estimated GFR < 60 mL/minute
 - Hypercalciuria ☰
 - Nephrolithiasis or nephrocalcinosis on imaging
 - Skeletal involvement
 - Reduced BMD (T-score ≤ -2.5 at any site)
 - Vertebral fracture

A. Adenoma → removed.

B. hyperplasia → subtotal resection (3 glands removed & the 4th is partially removed).

➤ Complication of the operation:

Hungry bone syndrome :

- Due to rapid mineralization of decalcified bone which occurs in patients with bone disease before surgery.
- It manifests as tetany due to hypocalcaemia & hypomagnesaemia.
- It can be prevented by giving Vit. D before the operation.
- Treatment of post operative tetany :
Ca I.V. + MgSo₄ (2-4 ml 50 % solution I V. infusion every 4 hour).

2. Medical ttt:

A. indication:

- Previous unsuccessful surgery. ‘
- Patient refusing the operation.
- Patient unfit for surgery.

B. drugs :

- Orthophosphate salts 1 - 3 gm 1/ day.
- Estrogen in ♀ (found to ↓ serum Ca in 1ry HPT).
- Dichloroethylene diphosphate (experimental).

Hypercalcaemic crisis

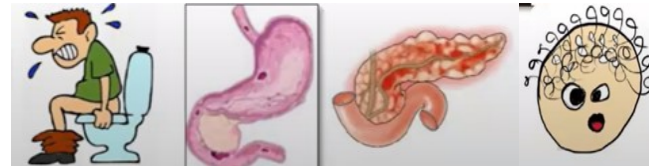
(serum Ca level > 13 mg %).

➤ **Causes :**

1. Iry HPT.
2. osteolytic carcinoma.
3. Multiple myeloma.

➤ **Manifestations :**

- Abdominal pain, vomiting, constipation.
- Polyuria, dehydration, weakness.
- Confusion, coma.
- CVS: Arrhythmia, Shortened QT interval.



Hypercalcaemic crisis

B. Treatment:

⚠ Caution

Start immediate hypercalcemia treatment for patients with a serum calcium level > 14 mg/dL (> 3.5 mmol/L).

1. **Rehydrate** the patient with IV normal saline: 4-6 L/day may be necessary.
2. **Diuretics:**
3. **Steroids** (Cortisone): is.
4. → ↓ serum Ca.

.

➤ **Treatment :**

1. Agents that ↑ urinary ca excretion :

- Hydration using isotonic saline. (the first goal)
- Frusemide → IV. 100 mg / hour. after patient is rehydrated. Should NEVER be administered without saline infusion, or the dehydration will be aggravated.
- Dialysis → for hypercalcaemia complicated by renal failure.

2. Agents that influence bone Ca fluxes :

Drug	Dose, mechanism of action & indication
mithramycin	inhibits bone resorption used particularly in malignant hypercalcaemia
PG inhibitors	Indomethacin or aspirin used in hypercalcemia due to osteolytic metastasis
Bisphosphonates	(Malignancies) as IV pamidronate: drug of choice. It inhibits the osteoclastic activity ↑ phosphate level → ↓ ca level
calcitonin	SC. It ↑ ca excretion in urine & ↓ ca release from bone
corticosteroids	Prednisone: ↓ bone resorption & ↓ intestinal absorption. The first line of treatment for hypercalcemia caused by vitamin D excess e.g. myeloma, lymphoma, sarcoidosis, but is NOT beneficial in hyperparathyroidism

Summary

Type	Ca	PO ₄	ALP	PTH	Management
Primary hyperparathyroidism	↑	↓	↑	↑/=	• Screen for MEN • Surgical removal of parathyroid glands is first line • If patients are unfit for surgery, medical therapy with alendronate and cinacalcet (calcimimetic agent that lowers calcium and PTH) may be attempted
Secondary hyperparathyroidism	↓	↑	↑	↑	• Medical therapy is first line – treat underlying cause • Treatment with CKD: 1. Phosphate binders such as sevelamer 2. Calcium supplementation 3. Vit D replacement
Tertiary hyperparathyroidism	↑	↑	↑	↑	• Surgical removal of the gland is first line • Cinacalcet may be used as adjunctive therapy

Differential Diagnosis

- From disorders causing:
 - Hypercalcaemia.
 - Generalized bone disease.

Causes of hypercalcaemia

1. Endocrinal:

- Hyperparathyroidism.
- Hyperthyroidism.
- Addison's disease.
- Acromegaly.

2. Malignant :

- Osteolytic metastasis: Local bone destruction by metastasis through prostaglandins or osteoclast activating factor (OAF).
- Multiple myeloma (OAF).
- Some lymphoma secrete 1, 25 (OH)₂ D₃.
- Ectopic: PTH secretion e. g. in bronchogenic carcinoma (PTH rP)

3.Iatrogenic:

- Hypervitaminosis
 - o D → treated with cortisone which blocks the action of vitamin D on the bone & intestine.
 - o A
- Milk alkali syndrome (Burnett syndrome).
- In patients with peptic ulcer ingesting excess amount of milk & alkalies for a long times.
 - o Early → mild alkalosis, hypercalcaemia & Hypercalciuria.
 - o Later → nephrocalcinosis, renal damage & uraemia which leads to → ↓ serum Ca & renal acidosis.

4.miscellaneous :

- Sarcoidosis, Paget's disease of bone (if the patient is immobilized)
- immobilization (as in paraplegia, fracture)
- idiopathic hypercalcaemia of infancy: 'idiopathic infantile hypercalcemia, IIH'
 - o Reversible form . due to ↑ Vit. D intake by the mother.
During lactation → failure to thrive, vomiting abdominal pain & constipation
 - o Irreversible form: due to ↑ intake of vit D by the mother during pregnancy → A.S. 'supravalvular', elfin facies, mental retardation & ends with renal failure.

causes of generalized bone disease.

1.Renal osteodystrophy : (see chronic renal failure).

2.Osteomalacia, Osteoporosis .

3.Paget's disease of bone : in which there is

- High alkaline phosphatase level.
- ↑ca level (if the patients is immobilized).
- X-ray → lytic areas in association with dense ones due to new bone formation.

PAGET'S DISEASE PATHOPHYSIOLOGY

- A chronic form of osteitis, most commonly affecting older people, causing thickening and hypertrophy of the long bones and deformity of the flat bones.
- The bone is excessively broken down (osteoclastic activity) and re-formed (osteoblastic activity).

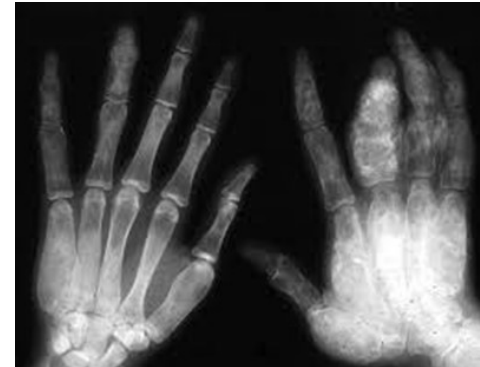


VIDEO



4. Me Cune Albright syndrome (rare):

- Fibrous dysplasia (unilateral).
- Areas of pigmentation.
- Precocious puberty (in ♀).



5. Osteoblastoma:

- Radiological appearance is similar to osteitis fibrosa cystica with no biochemical changes.

Hypercalcemia

Definition

Mild hypercalcemia: total serum Ca > 10.5 mg/dL or ionized Ca^{2+} > 5.25 mg/dL

Severe hypercalcemia: total Ca > 14 mg/dL or ionized Ca^{2+} > 10 mg/dL

Etiology

High PTH: hyperparathyroidism (1st–3rd degree), FHH

Low PTH: malignancies, granulomatous disorders

Other: medications (e.g., thiazide diuretics)

Diagnosis

Ionized Ca^{2+} or corrected serum Ca^{2+}

Serum intact PTH

ECG (e.g., QT interval shortening, J waves)

Bone evaluation

Treatment

Treat underlying cause

Ensure adequate hydration, reduce calcium intake

Severe hypercalcemia: rapidly lower calcium levels with IV 0.9% NaCl (+/- diuretics) and pharmacotherapy (e.g., calcitonin, bisphosphonates, denosumab)

Diminished muscle excitability

Cardiac arrhythmias
Muscle weakness, paresis

Gastrointestinal

Abdominal pain
Nausea and vomiting
Anorexia
Pancreatitis
Peptic ulcer disease
Constipation

Osteoarticular

Myalgias
Arthralgias
Bone pain
Fractures

Neuropsychiatric

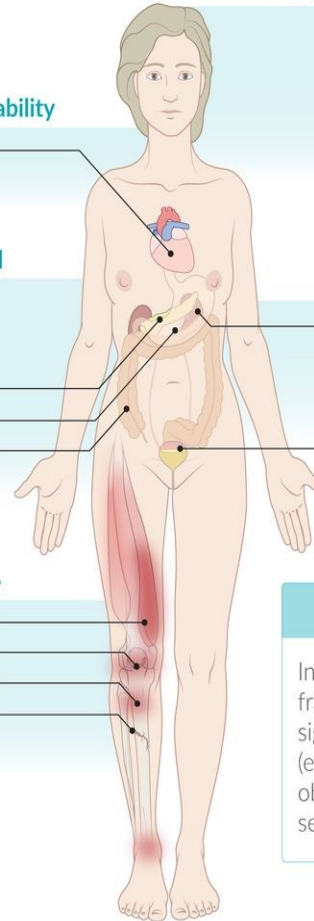
Anxiety
Depression
Fatigue
Cognitive dysfunction
Somnolence

Renal

Nephrolithiasis, nephrocalcinosis
(typically Ca^{2+} oxalate stones)
Polyuria and dehydration

Note

In the setting of hypercalcemia, fractures and bone pain may be signs of excessive bone resorption (e.g., due to malignancy), whereas obtundation and coma indicate severe hypercalcemia.



Case scenario :

Two hours after admission to the hospital for dialysis, a 63-year-old woman has severe pain in her lower back. The pain is 8 out of 10 in intensity and radiates down the legs. She underwent a laparoscopic cholecystectomy 4 years ago. She has hypertension, chronic kidney disease, type 2 diabetes mellitus, and major depressive disorder. She lives with her daughter and reports frequent fights with her at home. Her current medications include sertraline, insulin, enalapril, sodium bicarbonate, and sevelamer. She appears uncomfortable. Her temperature is 37.3°C (99.1°F), pulse is 102/min, respirations are 15/min, and blood pressure is 132/94 mm Hg. There is severe tenderness on palpation over the L2 vertebra; range of motion is limited. Neurologic examination shows no focal findings. Laboratory studies show:

Hemoglobin	10.1 g/dL
Leukocyte count	8,700/mm ³
Platelet count	130,000/mm ³
Serum	
Na ⁺	136 mEq/L
Cl ⁻	101 mEq/L
K ⁺	5.1 mEq/L
Calcium	8.2 mg/dL
Urea nitrogen	33 mg/dL
Glucose	190 mg/dL
Creatinine	3.8 mg/dL

An x-ray of the spine shows a wedge compression fracture of the L2 vertebra. Which of the following is the most likely explanation for these findings?

- A. Tertiary hyperparathyroidism
- B. Senile osteoporosis
- C. Secondary hyperparathyroidism
- D. Postmenopausal osteoporosis
- E. Elder abuse
- F. Pseudohypoparathyroidism
- G. Primary hyperparathyroidism

Summary

I just need
the main ideas



References

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Thank You